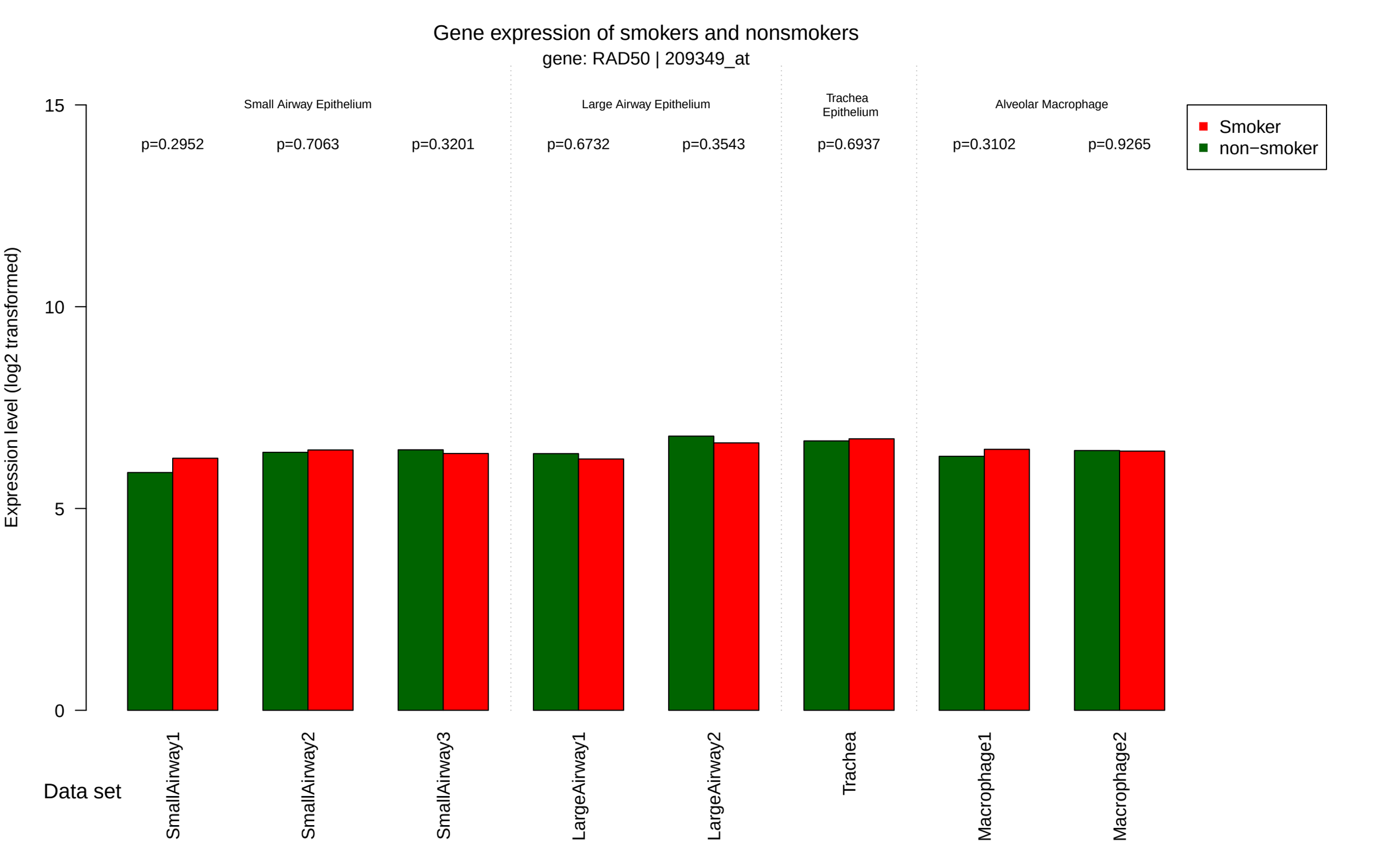
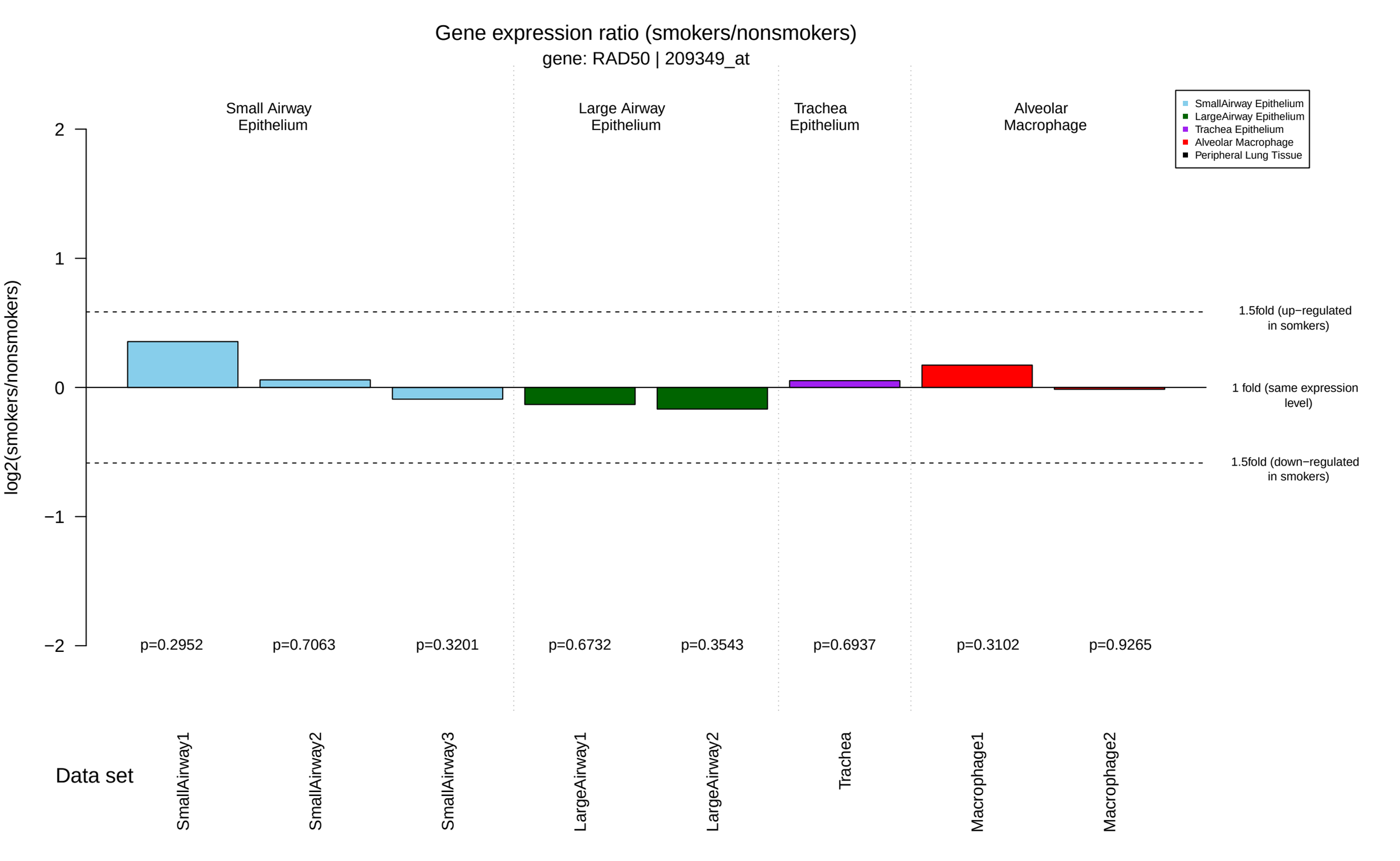
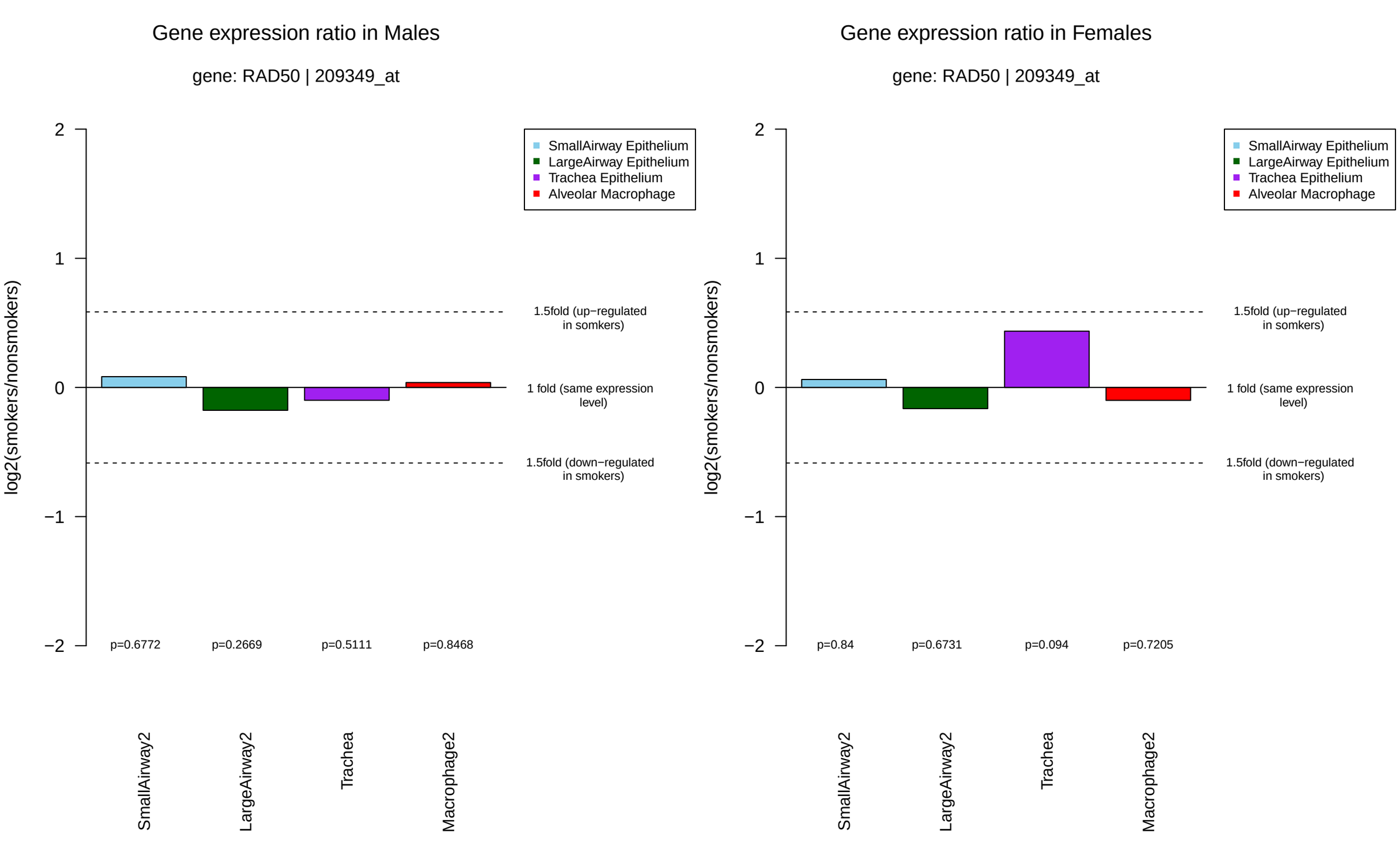
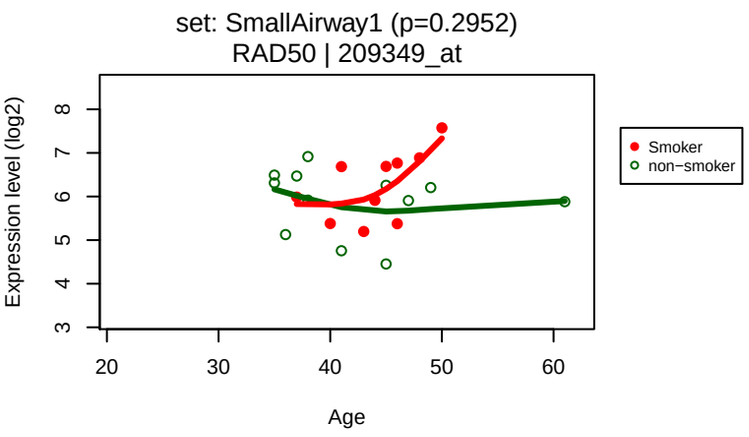
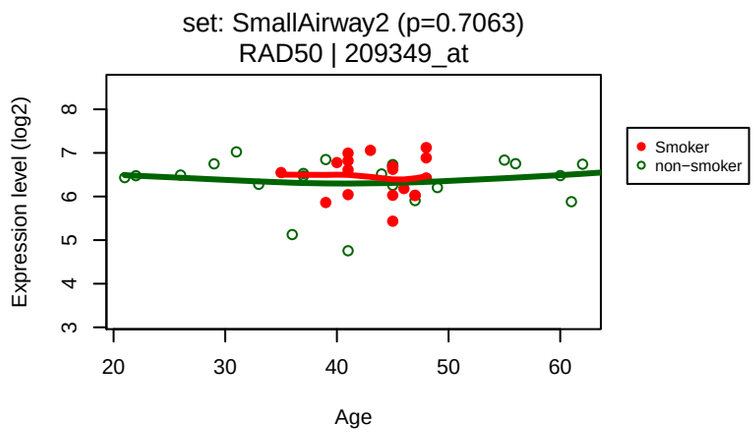
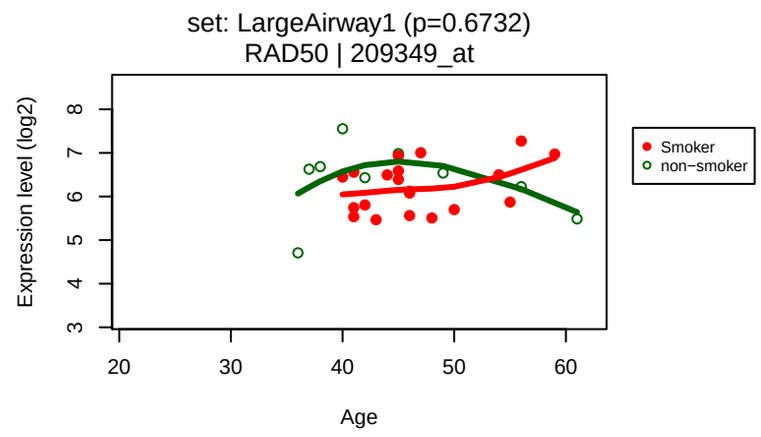

Gene expression of smokers and nonsmokers
gene: RAD50 | 209349_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2952
p=0.7063
p=0.3201
p=0.6732
p=0.3543
p=0.6937
p=0.3102
p=0.9265
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

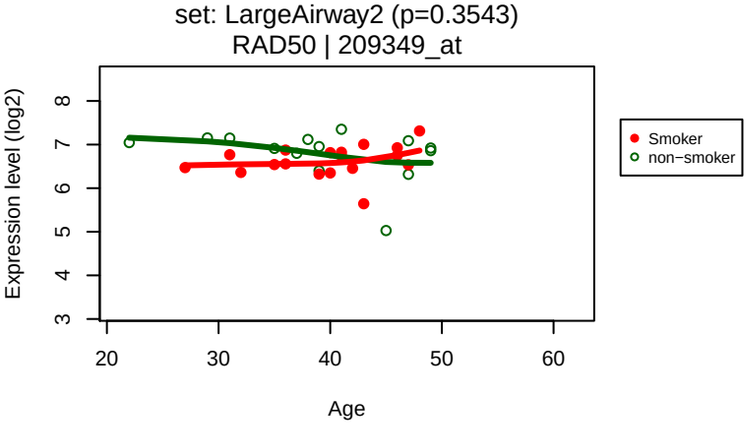

Gene expression ratio (smokers/nonsmokers)
gene: RAD50 | 209349_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2952
p=0.7063
p=0.3201
p=0.6732
p=0.3543
p=0.6937
p=0.3102
p=0.9265
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

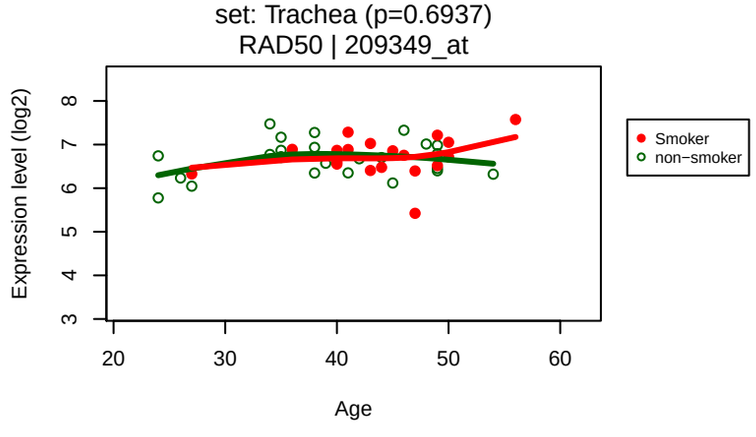

Gene expression ratio in Males
gene: RAD50 | 209349_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6772
p=0.2669
p=0.5111
p=0.8468
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD50 | 209349_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.84
p=0.6731
p=0.094
p=0.7205
SmallAirway2
LargeAirway2
Trachea
Macrophage2

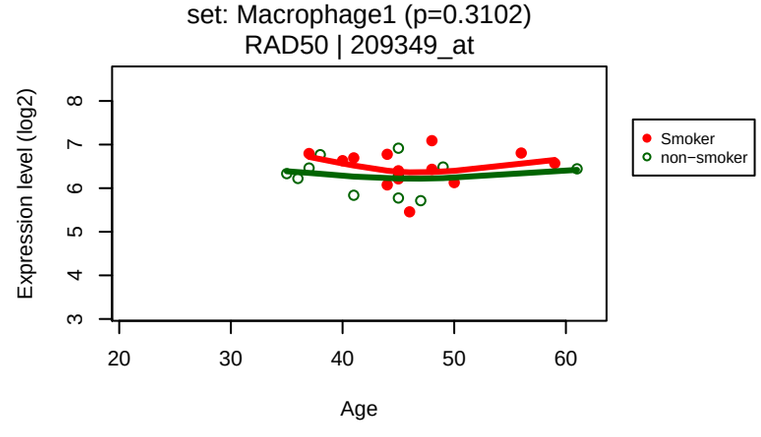

set: SmallAirway1 (p=0.2952)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker

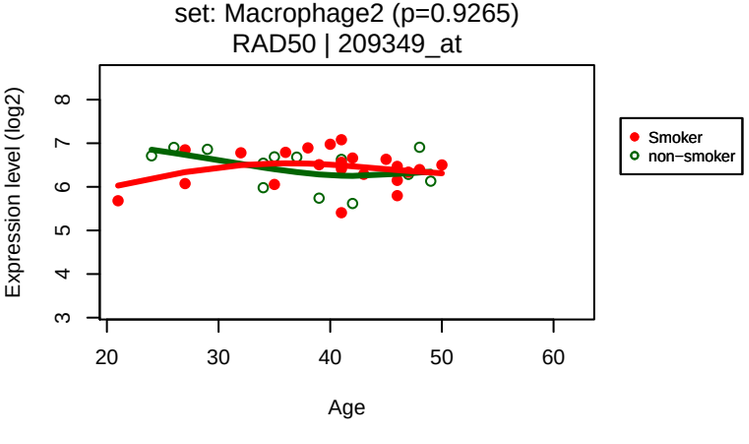

set: SmallAirway2 (p=0.7063)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker

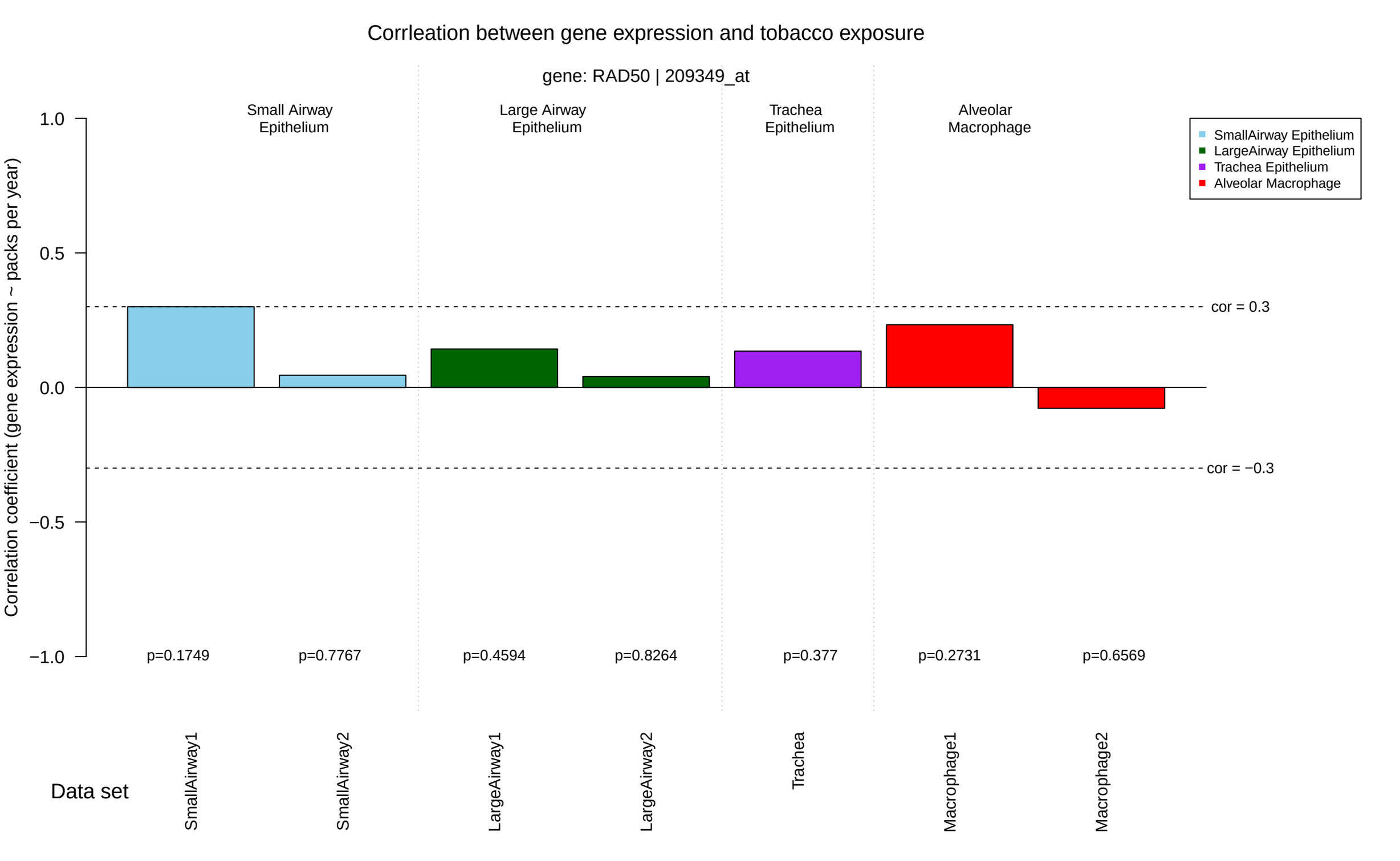

set: LargeAirway1 (p=0.6732)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3543)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6937)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3102)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9265)
RAD50 | 209349_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD50 | 209349_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1749
p=0.7767
p=0.4594
p=0.8264
p=0.377
p=0.2731
p=0.6569
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2